MANCHESTER 1824
The University of Manchester

Understanding the Earth

Week 4 But which way *is* up?

The aim of this week's workshop is:

- to what features in the rocks can be used to determine which way was originally up

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Week 4 But which way *is* up?

On successful completion of the workshop you will be able to:

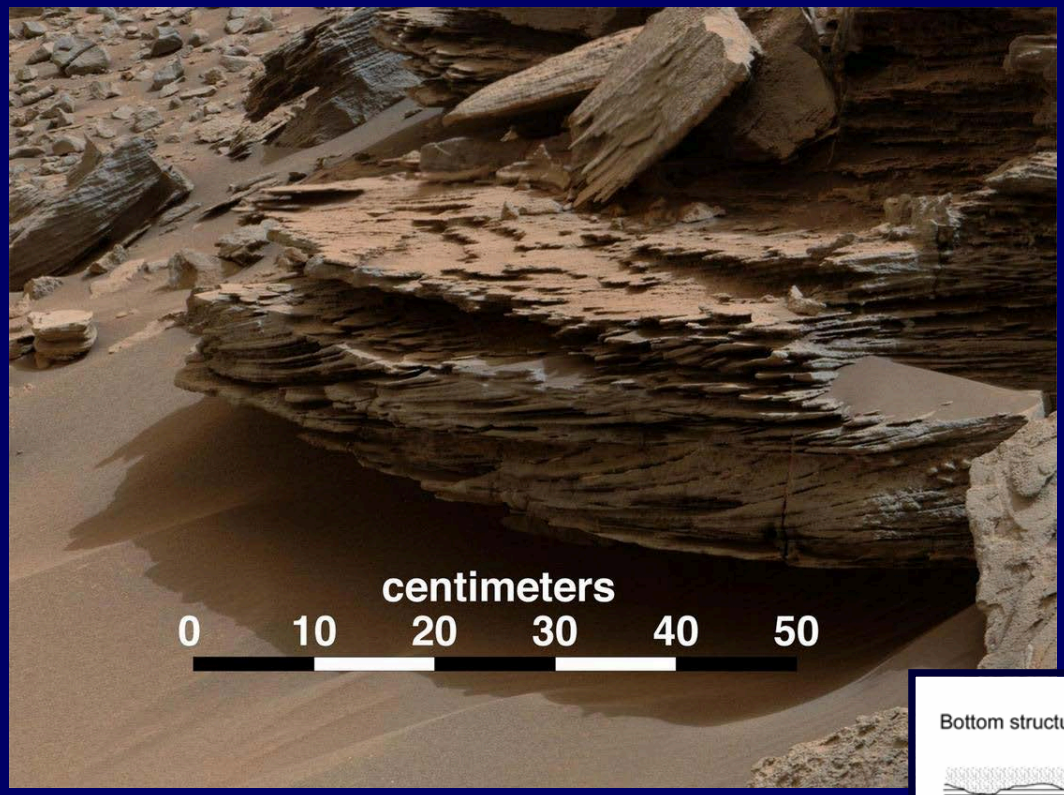
- determine ‘way up’ using features in the rock
- use a younging symbol to show the way up of a sequence of beds

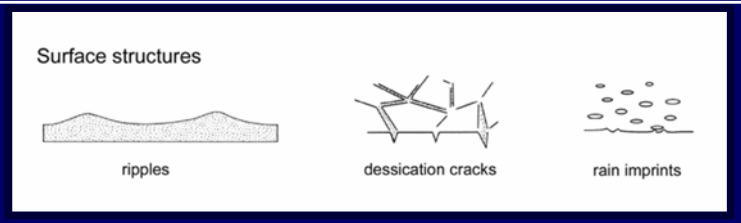
Way-up structures: Significance

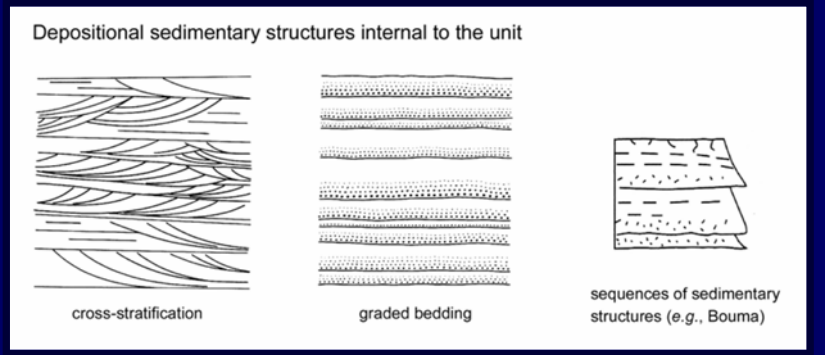


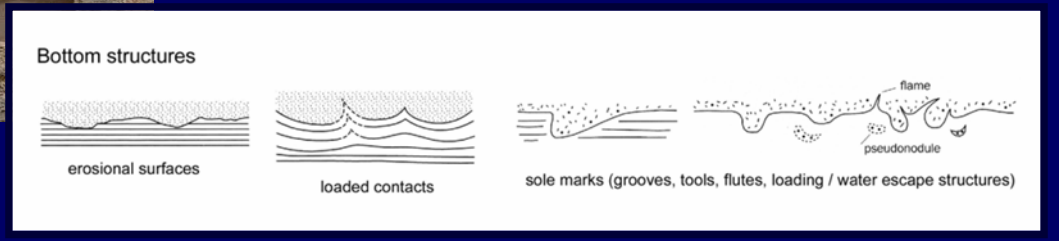
Steeply dipping Devonian units – which way do they young?

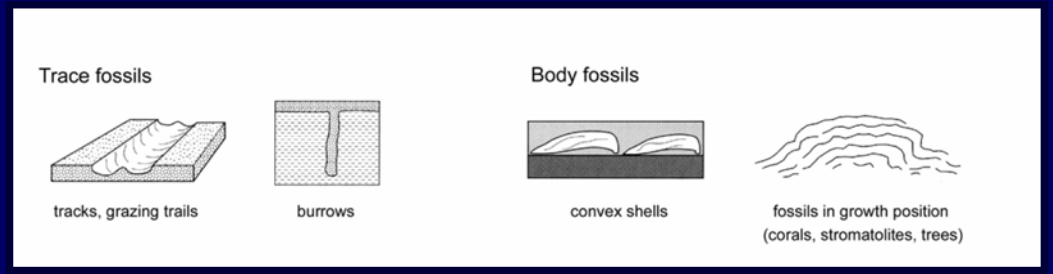
Way-up structures: Examples

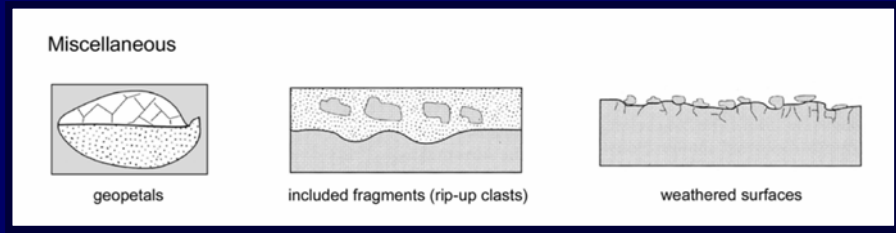












Way-up structures – ripples

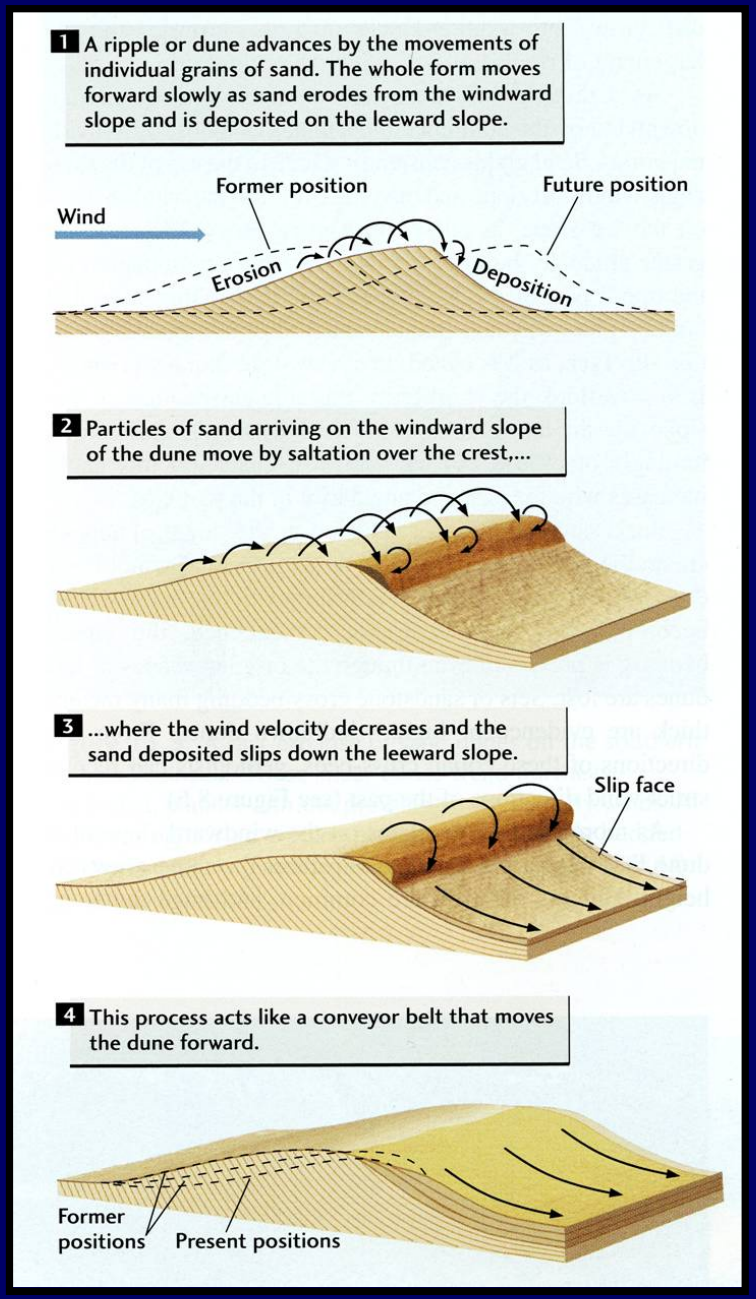


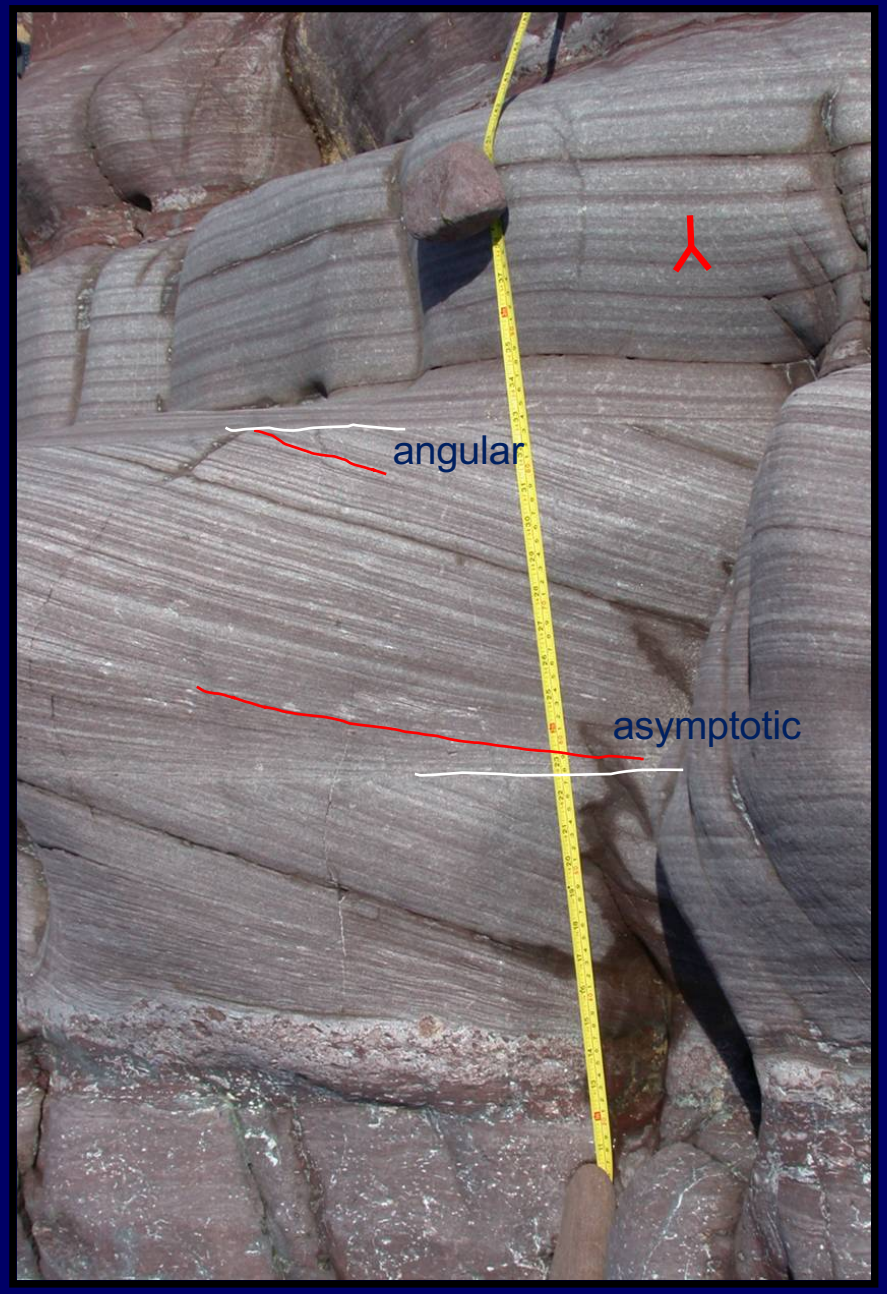






Way-up structures – cross-bedding



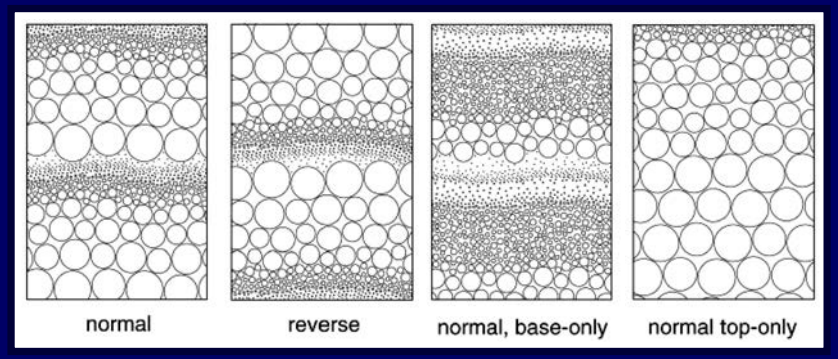


Way-up structures – cross-bedding 2



Way-up structures – graded bedding





Normal graded bedding is widely used for way-up, but be alert to the possibility of reverse grading

Way-up structures – sole marks on underside of beds



Flutes



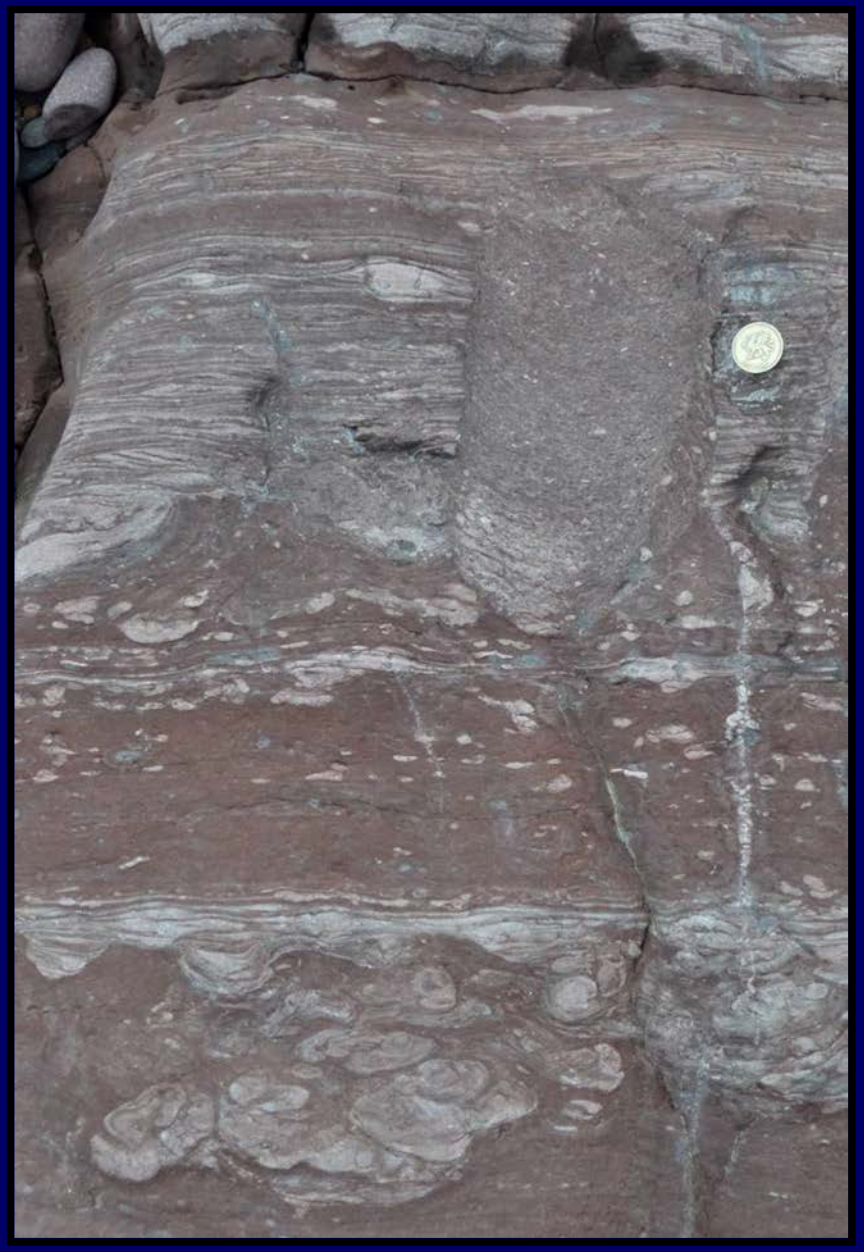
Furrows

Way-up structures – loading structures



Sand dumped rapidly on ‘sloppy’ muds leads to a variety loading structures and . . .

Practice



How many features and principles can you identify?

Use a younging symbol to show which is the correct way up of the sequence

Can you do what we set out to do?

On successful completion of the workshop you will be able to:

- determine ‘way up’ using features in the rock
- use a younging symbol to show the way up of a sequence of beds